

Coupled nTop and SUAVE Conceptual Sizing

A solid-propellant rocket vehicle generator, and what the loop caught

Prepared with nTop Automate 5.53.2 / 5.54.0 and SUAVE 2.5.2. All geometry is authored programmatically. Report generated from the run artefacts under runs/SV-1.

Summary

This report describes a conceptual sizing loop for a solid-propellant rocket vehicle. SUAVE does the physics. nTop does the geometry. The two are coupled: nTop measures the solid it builds, and those measurements go back into the mass and aerodynamic models. The loop then repeats until the launch mass stops changing.

The loop found a design that meets all ten constraints. It weighs 554.3 kg. It flies 189.5 km. It arrives at Mach 1.66.

The result matters less than what the loop caught. The requirement set was self-contradictory. One requirement was impossible for this vehicle class. One motor transition needs hardware that does not exist. Five defects in the code and in the study method would have corrupted the answer. Section 1 gives each finding.

The vehicle is invented for this demonstration. Its requirements match no real programme. Section 6 lists every limitation and every value that is a guess.

Quantity	Value	Requirement	Status
Launch mass	554.3 kg	≤ 1100 kg	met
Range	189.5 km	≥ 185 km	met
Mach at impact	1.66	≥ 1.50	met
Maximum dynamic pressure	195.1 kPa	≤ 200 kPa	met, active
Body diameter	0.350 m	≤ 0.45 m	met
Overall length	3.60 m	≤ 4.20 m	met

Table 1. Headline result. Dynamic pressure is the active constraint. Source: runs/SV-1/converged/point_ntop.json.

1. What the loop caught

A sizing loop earns its cost when it finds errors that inspection misses. This one found five. Each was found by running the constraint set, not by reading the specification.

1.1 The requirement set contradicted itself

Two requirements could not both hold. R6 asks for Mach 1.50 at impact. The impact is at sea level. The structural limit on dynamic pressure was first written as 90 kPa. Mach 1.50 at sea level is itself 159.6 kPa.

Step	Value
Sea-level speed of sound, US Standard 1976	340.29 m/s
Sea-level density	1.2250 kg/m ³
Speed required by R6	510.4 m/s
Dynamic pressure at that speed	159.6 kPa
Mach permitted by a 90 kPa limit	1.126

Table 2. The R6 against $q_{\max}$ contradiction. A 90 kPa limit caps impact at Mach 1.13, so R6 was unreachable by definition.

The limit was wrong, not the physics. R6 sets its own floor of 159.6 kPa. The limit is now 200 kPa. That clears the floor with about 25 percent margin. It also agrees with the sea-level supersonic pressures quoted for this vehicle class in Fleeman, Tactical Missile Design, 2nd edition, Chapter 3.

1.2 R6 is impossible without thrust in the endgame

An unpowered dive cannot reach Mach 1.50 at sea level. The reason is not the design. The dive reaches terminal velocity, where drag balances weight. No dive angle and no propellant loading changes that.

Terminal dive angle [deg]	Range [km]	Mach at impact
-25	245.4	0.673
-35	237.1	0.802
-50	230.5	0.905
-70	225.5	0.961
-89	222.2	0.974

Table 3. Dive-angle sweep at fixed propellant. Impact Mach approaches an asymptote just below 1. Source: `evidence.py`, `unpowered_dive_sweep`.

Closed form confirms it. At the burnout mass of 203.1 kg, the sea-level terminal velocity is $\sqrt{2mg/(\rho S C_D)}$. Solving it together with the drag model gives 318.3 m/s. That is Mach 0.935. To reach Mach 1.50 unpowered the total drag coefficient would have to fall to 0.130, which this configuration cannot approach.

The fix is a terminal boost. Real supersonic terminal-phase vehicles carry thrust into the endgame. The motor now has a third phase. It ignites during the dive. Its propellant comes out of the sustain charge, not on top of it.

Terminal propellant [kg]	Range [km]	Mach at impact	q_max [kPa]
0	237.1	0.802	148.8
10	216.3	1.160	148.8
20	208.0	1.349	148.8
30	199.6	1.513	162.3
40	191.4	1.665	196.6
50	183.1	1.819	234.6
60	174.6	2.023	290.2

Table 4. Terminal-boost trade at constant total propellant of 212 kg. Range falls by about 0.97 km for each kilogram moved into the terminal charge.

Terminal boost moved the active constraint. A powered dive accelerates into thick air. Dynamic pressure therefore climbs steeply with terminal propellant. The converged design now sits against the 200 kPa limit at 195.1 kPa, not against the Mach limit. The 90 kPa error in section 1.1 had to be corrected before this design could exist.

1.3 One motor transition needs hardware that does not exist

Transition	From [mm^2]	To [mm^2]	Direction	Credible
boost to sustain	4,024	1,350	decrease	NO
sustain to terminal	1,350	1,350	unchanged	yes

Table 5. Throat transitions. The boost-to-sustain step needs the throat to shrink. The model reports the mechanism it would need: no insert mechanism shrinks a throat; needs a separate smaller sustainer nozzle or a jettisoned tandem booster.

No mechanism shrinks a rocket nozzle throat. An ejected insert only enlarges one. Erosion also only enlarges one. A smaller effective throat after boost needs a separate sustainer nozzle in the same aft closure, or a jettisoned tandem booster. Neither is a single-throat motor. The model reports this transition as not credible and names the hardware it would need. A test asserts the report, so the statement cannot decay into prose.

The terminal pulse needs no new hardware. It shares the sustain throat. Its chamber pressure follows from its own burning area.

1.4 Bottom-up motor inert mass is optimistic

The bottom-up sum of case, insulation, nozzle and igniter gives 21.0 kg. That is a propellant mass fraction of 0.942. Real tactical motors achieve 0.80 to 0.92. The sum is incomplete: it omits thrust skirts, case joints, closure hardware, the aft attachment ring, the nozzle ablative liner and the exit cone.

The correlation band governs. The shortfall of 12.9 kg appears in the mass statement as its own line item, named Motor hardware not modelled. It is not absorbed into another line. A reader can see exactly how much of the motor mass is a correlation.

1.5 Five defects caught by the test suite

Defect	Effect if unfixed
The mass statement did not count the terminal propellant.	Launch mass was 28 kg light. Fixed, then hardened: one shared list of propellant items now drives every total, so a new burn phase cannot be dropped again.
The motor case was sized from the boost and sustain pressures only.	A high-pressure terminal pulse rode for free. At 14 kN terminal thrust the terminal chamber pressure exceeds boost pressure. The wall grows from 1.20 to 1.72 mm.
The convergence gate needed two iterations.	No point could report convergence at a one-iteration budget. Every trade-study row run at that budget was silently marked as failed.
The trade-study axes did not bracket the converged design.	A 75-node factorial reported an empty feasible region. The region exists. The axes were re-centred on the sized point.
The results writer took its column header from the first row.	A sample that failed before the constraints were built has no margin columns, so the export crashed. The header is now the union of all rows.

Table 6. Defects found by the tests, and what each would have cost.

2. The converged design



Figure 1. The converged SV-1, rendered from the STL that nTop exported. Tangent-ogive nose, cylindrical mid-body, conical boattail, four cruciform tail fins. The geometry is authored programmatically; no one opened the nTop graphical interface.

Parameter	Value	Unit
Body diameter	0.350	m
Overall length	3.600	m
Nose fineness, nose length over diameter	3.40	-
Nose length	1.190	m
Body fineness, length over diameter	10.29	-
Reference area	0.0962	m ²
Wall thickness	3.00	mm
Base diameter	0.300	m
Fin count	4	-
Fin exposed semi-span	0.230	m
Fin root chord	0.420	m
Fin taper ratio	0.45	-
Fin leading-edge sweep	45.0	deg
Boost propellant	130.0	kg
Sustain propellant	172.0	kg
Terminal propellant	40.0	kg
Boost thrust	45.0	kN

Parameter	Value	Unit
Terminal thrust	8.0	kN
Chamber pressure	7.00	MPa
Nozzle area ratio	8.0	-

Table 7. The converged design vector.

2.1 Mass statement

The launch mass is 554.3 kg. The centre of gravity is 1.917 m aft of the nose tip. At burnout the mass is 212.3 kg and the centre of gravity moves forward to 1.380 m. That forward shift is what erodes static margin through the burn.

Item	Mass [kg]	Percent	Station [m]	Provenance
Propellant, sustain	172.00	31.0	2.553	analytic
Propellant, boost	130.00	23.5	1.537	analytic
Warhead	90.00	16.2	0.825	requirement
Airframe structure and fins	41.07	7.4	2.285	ntop measured
Propellant, terminal	40.00	7.2	3.265	analytic
Dry-mass contingency	18.66	3.4	1.789	correlation
Guidance, seeker, actuation	15.00	2.7	0.425	requirement
Radome	13.78	2.5	0.535	analytic
Motor hardware not modelled	13.15	2.4	2.250	correlation
Motor insulation	10.51	1.9	2.250	analytic
Motor case	5.60	1.0	2.250	analytic
Nozzle assembly	2.85	0.5	3.520	correlation
Igniter	1.71	0.3	1.150	analytic
TOTAL	554.34	100.0	1.917	

Table 8. Group-weight statement. Provenance says where each mass came from. Only 7.4 percent of the launch mass is measured by nTop. The payload and the propellant are requirements and correlations, not geometry, so they cannot be measured.

Launch mass 554.3 kg, CG 1.917 m from the nose tip, burnout mass 212.3 kg. nTop-measured share 7.4 %.
Source: runs/SV-1/converged/point_ntop.json.

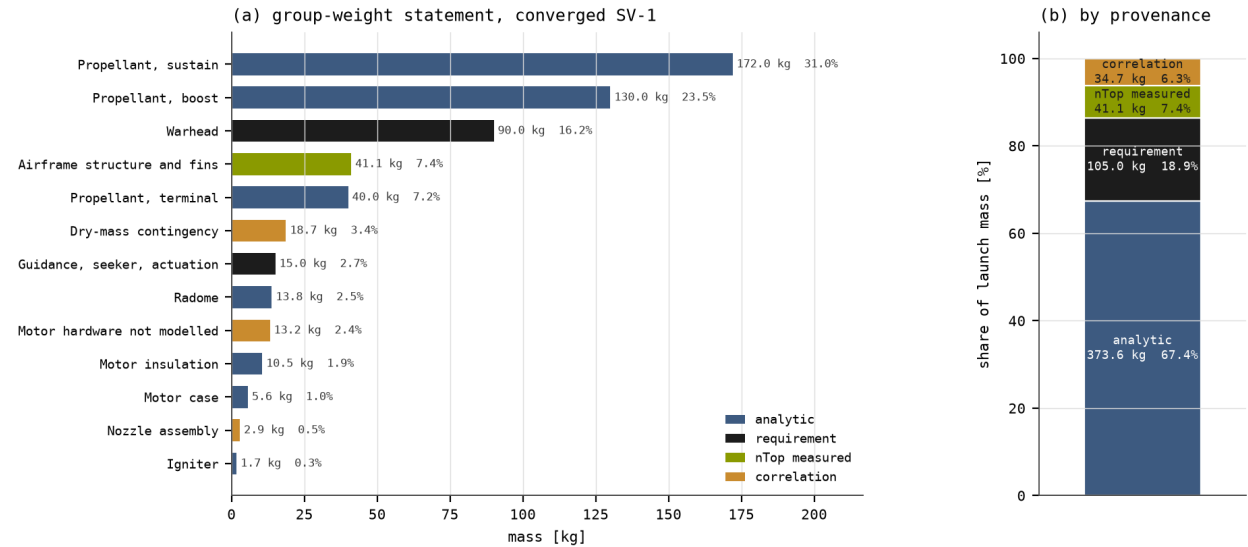


Figure 2. The mass statement, coloured by provenance. The measured fraction is small because propellant and payload dominate the mass and neither comes from geometry.

2.2 Constraints

Constraint	Value	Sense	Limit	Margin	Status
R3 range	189,480.140	>=	185,000.000	+2.4	met
R6 terminal Mach	1.659	>=	1.500	+10.6	met
R7 diameter	0.350	<=	0.450	+22.2	met
R8 length	3.600	<=	4.200	+14.3	met
R9 launch mass	554.344	<=	1,100.000	+49.6	met
R10 static margin	1.227	>=	1.000	+22.7	met
R11 fin span	0.810	<=	0.900	+10.0	met
q_max	195,100.299	<=	200,000.000	+2.4	met
grain L/D lower	7.405	>=	1.000	+640.5	met
grain L/D upper	7.405	<=	8.000	+7.4	met

Table 9. All ten constraints, with margin as a percentage of each limit. Dynamic pressure has the smallest margin, so it is the active constraint.

Bars are the nTop-coupled result; the red ticks are the same design vector with analytic geometry.
The shaded band marks margins below 5 percent. Source: runs/SV-1/converged/point_ntop.json.

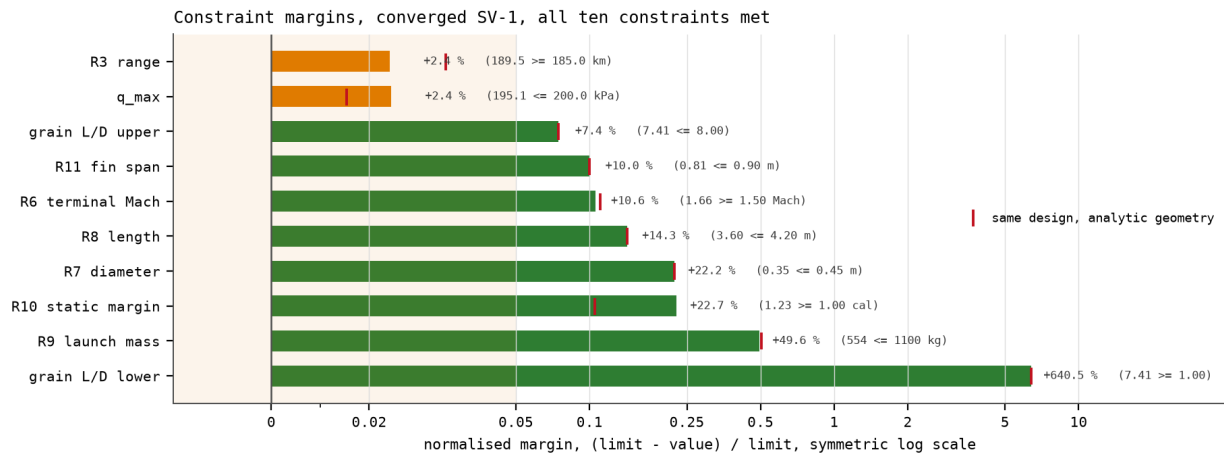


Figure 3. Normalised constraint margins for the converged design. Dynamic pressure binds.

2.3 Trajectory

The mission has five phases: separation, boost, sustain, dive and terminal boost. The flight lasts 308.9 s. It covers 189.5 km. The rocket arrives at Mach 1.66.

The integrator reported: ground impact; sustain phase could not hold constant Mach at constant altitude: the motor was short by up to 5443 N. The trajectory shows what the motor actually delivered.; terminal dive entered at t = 258.0 s; terminal boost lit at t = 297.9 s for 12.7 s; terminal boost was still burning 1.7 s of propellant worth at impact; that propellant was wasted

SV-1 trajectory
range 189.5 km terminal M 1.66 q_max 195.1 kPa converged True

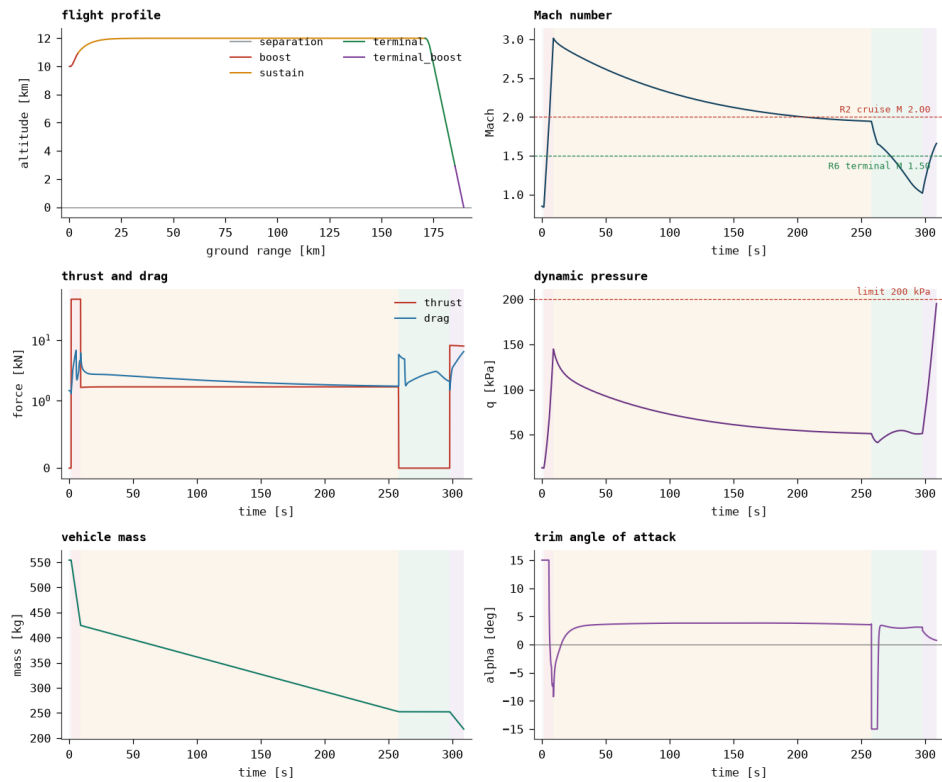


Figure 4. The converged trajectory. Phases are shaded. The terminal boost is the last phase, and it is what raises the impact Mach above the requirement.

3. The nTop coupling

This section is the point of the exercise. nTop is not a downstream renderer here. It measures the solid it builds, and the measurements change the answer.

3.1 The notebook is authored programmatically

An nTop notebook is a binary container. It cannot be written by hand. The notebook is therefore emitted as recipe JSON and converted with an undocumented ntopcl subcommand:

```
ntopcl convert <recipe.json> <out.ntop> --dev-blocks-on=True
ntopcl exportjson <in.ntop> <out.json> --ext --dev-blocks-on=True
```

The two commands round-trip. The recipe is built against a universe of 853 block signatures. All fifteen geometry variables are real notebook inputs. One notebook therefore serves every design point: the loop converts once, then runs many times with different input JSON. This matters because conversion is expensive. Conversion evaluates the notebook, exports included.

The outer mould line is one closed profile revolved through 360 degrees. The fins are extruded plates, patterned about the axis. The airframe is then hollowed with an inward offset, and the internal bays are subtracted, so the cavity is a real volume rather than an assumed fraction.

3.2 Measurement accuracy

Quantity	nTop	Closed form	Difference
volume total	0.290635	0.290689	-0.018
area wetted body	3.523487	3.519386	+0.117
area wetted fins	0.619175	0.560280	+10.512
area base	0.070686	0.070686	+0.000

Table 10. nTop measurements against independent closed-form geometry. The closed form is separately tested against an exact hemisphere, so it is a trustworthy reference. The closed form counts two sides of each exposed panel only. The nTop plate has edges as well, so a positive error is expected.

nTop reports a structure mass of 41.07 kg for the hollow airframe and fins. A solid billet of the same envelope would weigh 817 kg. The structure is therefore a real hollow shell, not a filled body. The measured centre of gravity is 2.2846 m aft of the nose tip, and it sits 0.3 mm off the axis. That small offset is discretisation, not asymmetry, so tests check it against a tolerance and never against zero.

S(x) from the nTop notebook against the closed-form outer mould line plus plate fins.
Source: runs/SV-1/converged/measurements.json.

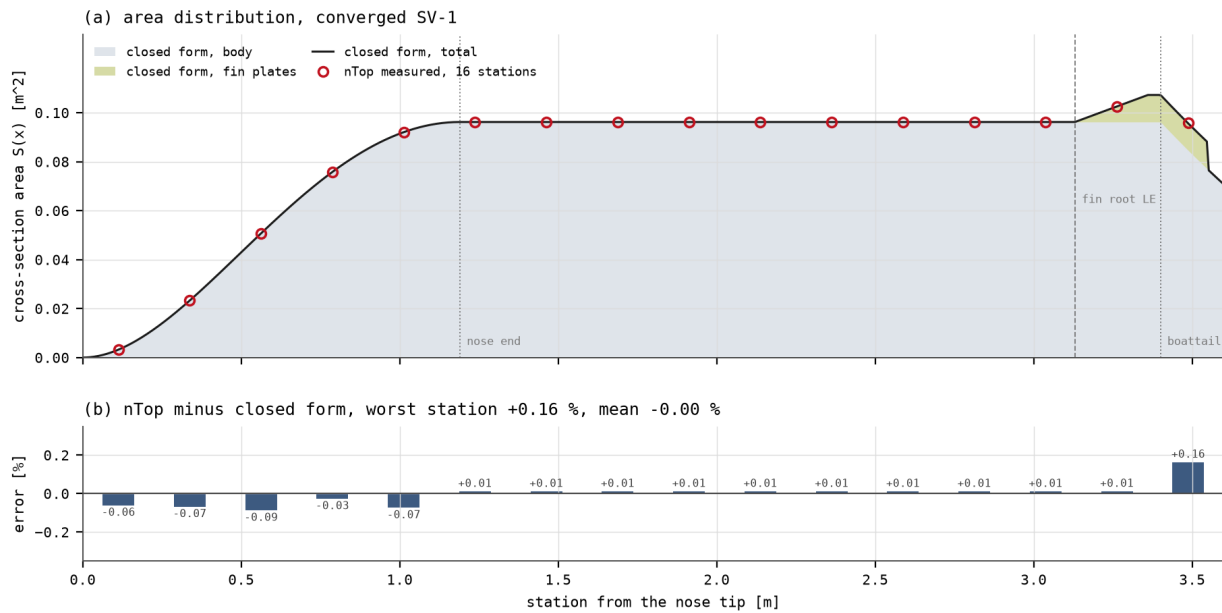


Figure 5. Cross-section area against station, measured in nTop by sectioning the solid at 16 stations, against the closed-form outer mould line.

3.3 The coupling changes the answer

Quantity	Analytic geometry	nTop-measured	Change
Launch mass [kg]	545.1	554.3	+9.3
Range [km]	191.6	189.5	-2.1
Mach at impact	1.67	1.66	-0.01
Maximum dynamic pressure [kPa]	196.9	195.1	-1.8
Centre of gravity [m]	1.932	1.917	-0.015

Table 11. The same design vector, sized with analytic geometry and with nTop geometry. This is a real coupling effect, not numerical noise.

The measured airframe is heavier than the analytic shell estimate. Launch mass therefore rises by 9.3 kg. A heavier rocket flies 2.1 km less far. Both answers meet every requirement, so the coupling does not change the verdict here. It does change the numbers, and it would change the verdict on a design with less margin.

One measurement call takes about 36 s. Exports are off by default. Nothing that is measured comes off the exported mesh, so the loop loses no information when exports are off. Exports run for the converged design only.

4. Trade study

Two studies were run. A full factorial of 45 nodes covers body diameter, sustain propellant and nose fineness. A Latin hypercube of 40 samples covers eight variables. The hypercube uses a seeded generator, so it repeats exactly on any machine. Failed samples are recorded, never discarded: a study that drops its failures reports a feasible region that is too large.

Only 3 of 45 factorial nodes are feasible, and 1 of 40 hypercube samples. The feasible region is narrow. That is the single most useful output of the study.

SV-1 trade study: 45-node full factorial over D, sustain propellant and nose fineness.
3 of 45 nodes are feasible. Source: runs/SV-1/dae/grid.csv.

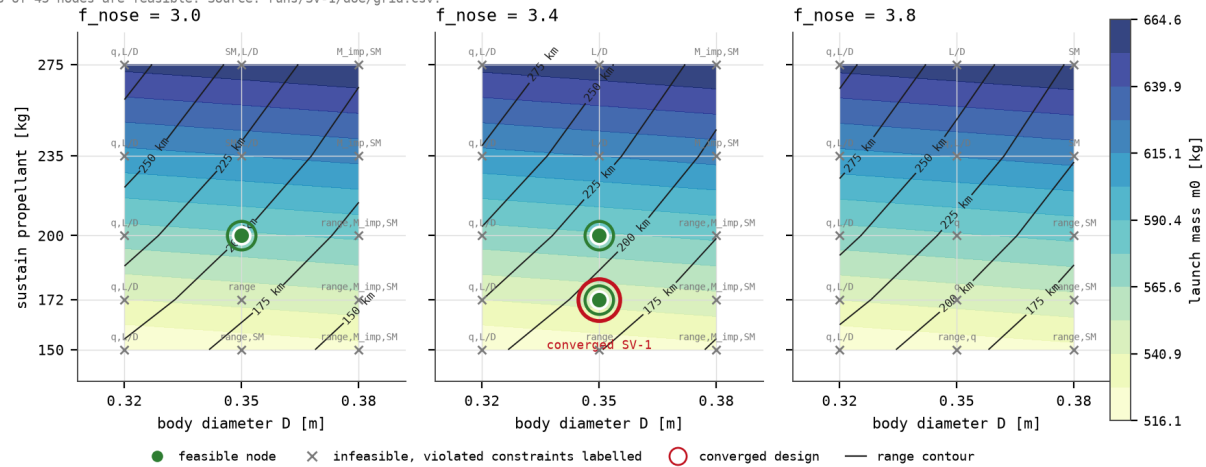


Figure 6. Launch mass and range over body diameter and sustain propellant, at three nose finenesses. Feasible nodes are filled. Infeasible nodes carry the constraints they violate. The converged design is ringed.

D [m]	Sustain [kg]	Nose fineness	Mass [kg]	Range [km]	Mach	q [kPa]
0.350	172	3.40	545.1	191.4	1.66	197
0.350	200	3.00	576.0	200.1	1.61	184
0.350	200	3.40	576.3	210.2	1.66	195

Table 12. Every feasible node of the factorial, lightest first. The lightest is the design the search returned. That is an independent check on the search, because the factorial did not use the search.

4.1 Sensitivity

Sensitivity uses Spearman rank correlation, not Pearson. The responses are monotone in the design variables but not linear, so rank correlation measures the association without assuming a shape. Only converged samples are counted.

Variable	Launch mass	Range	Impact Mach	q _{max}
D	+0.110	-0.785	-0.758	-0.661
L _{total}	+0.134	+0.128	-0.141	-0.053
f _{nose}	-0.037	+0.002	-0.055	+0.105

Variable	Launch mass	Range	Impact Mach	q_max
m_p_sustain	+0.723	+0.580	-0.026	-0.206
m_p_boost	+0.556	+0.114	-0.296	-0.020
m_p_terminal	+0.018	-0.137	+0.617	+0.335
b_fin	-0.039	-0.177	-0.129	-0.105
c_r_fin	-0.355	-0.208	-0.006	+0.084

Table 13. Spearman rank correlation from the Latin hypercube.

Sensitivity from the 40-sample Latin hypercube over eight variables (0 failed samples). Source: runs/SV-1/dae/sensitivity.json.

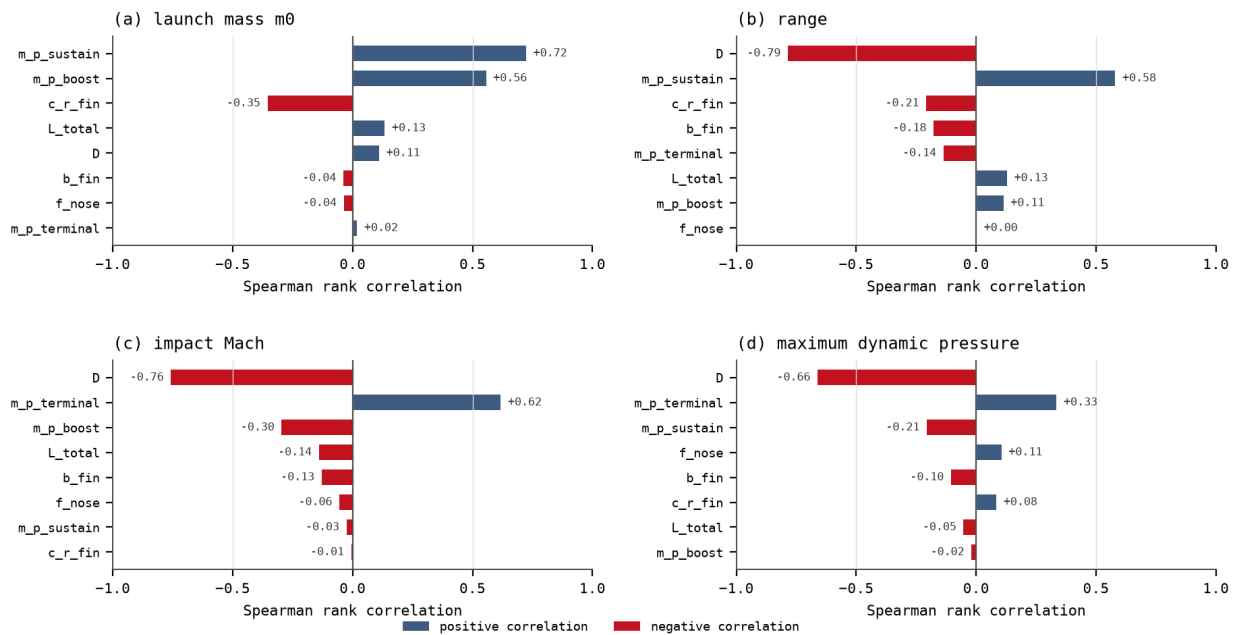


Figure 7. Rank-correlation sensitivity, grouped by response.

The rankings agree with the physics. Sustain propellant drives launch mass and range. Terminal propellant drives impact Mach, and it is the second strongest driver of dynamic pressure. Body diameter drives range, impact Mach and dynamic pressure, all negatively, because a smaller body has less drag.

5. Verification

296 automated tests pass. The physics is checked against closed-form answers where they exist, and against published measurements where they do not.

5.1 The trajectory integrator

Case	Relative error
Vacuum ballistic range against the closed-form parabola	2.0e-14
Vacuum ballistic apogee	2.1e-15
Burnout speed against Tsiolkovsky less gravity loss	1.7e-13
Steady terminal velocity against $\sqrt{2mg/(\rho S CD)}$	8.7e-10
Specific-energy drift over 100 s with no thrust and no drag	6.5e-15

Table 14. Integrator validation. Halving the step reduces the error by 15.93, 16.01, 15.76, against the factor of 16 that fourth-order Runge-Kutta requires. The integrator is therefore correct to machine precision on every case with an analytic answer.

5.2 The aerodynamic model

The build-up is validated against free-flight measurements of the Army-Navy Basic Finner. The reference is A. D. Dupuis and W. Hathaway, Aeroballistic Range Tests of the Basic Finner Reference Projectile at Supersonic Velocities, Defence Research Establishment Valcartier, DREV-TM-9703, 1997, Table VII. It gives 23 shots between Mach 1.06 and Mach 4.47.

Quantity	Shots	Mean bias [%]	Worst shot [%]
Zero-lift drag coefficient	17	-14.6	+22.9
Normal-force slope	13	-10.7	+29.0
Centre of pressure	13	+2.0	+11.4

Table 15. Aerodynamic validation against Basic Finner free-flight data.

The drag bias is systematic, not random. The model runs about 14.6 percent low. Three causes are known: fin trailing-edge base drag is not modelled, fin-body junction interference is not modelled, and the real fin section is conical rather than the double wedge the wave drag assumes. A low drag coefficient overpredicts range, which is the wrong direction for a sizing tool. Drag is therefore scaled by 1.171 in the sizing loop. The correction is applied at the loop boundary, never inside the aerodynamic model, so the model always reports what its physics gives.

Centre of pressure and normal-force slope are left uncorrected. Their agreement is good enough that a correction is not justified.

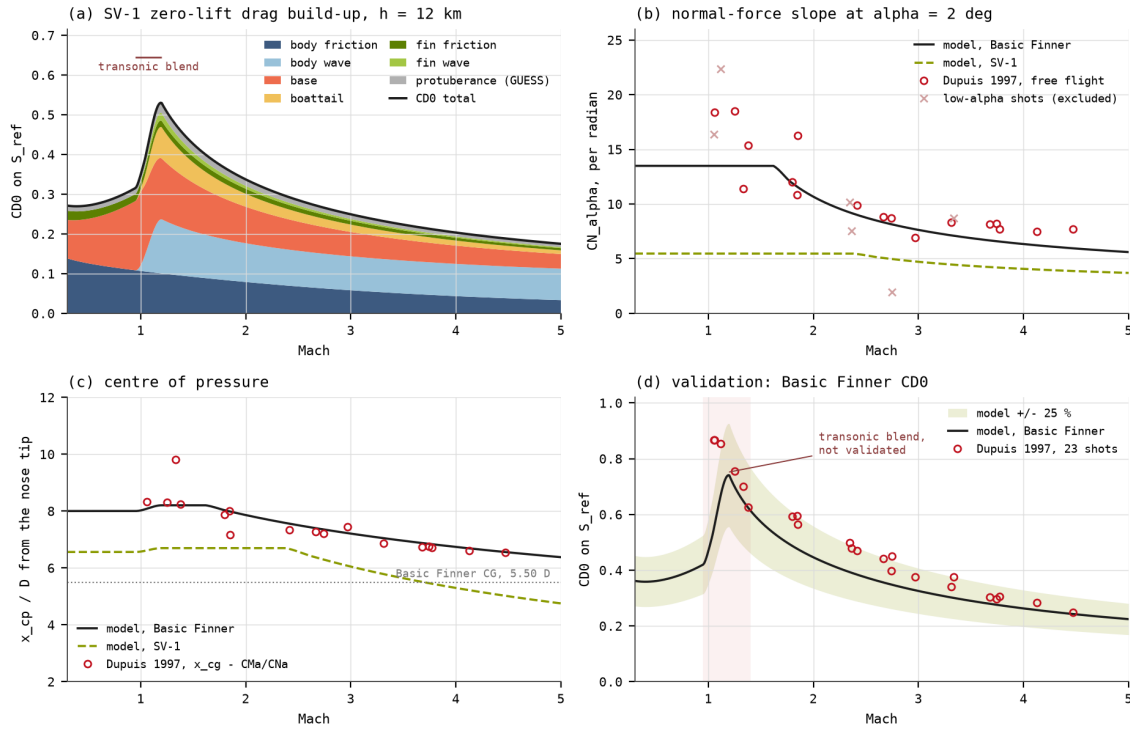


Figure 8. Drag build-up, normal-force slope and centre of pressure against Mach, with the Basic Finner measurements overlaid. The transonic region is a blend, not physics, and is excluded from validation.

5.3 The motor and the geometry

The nozzle thrust coefficient reproduces published isentropic tables to better than 0.1 percent. Total impulse equals specific impulse times standard gravity times propellant mass, summed over all three burn phases, and a separate integral checks propellant-mass conservation over the whole timeline.

The geometry pipeline was proved on a sphere before the rocket was attempted. A 25 mm sphere gave a volume error of 0.0104 percent from the notebook's own mass properties, and 0.169 percent from the exported mesh. The notebook measurements are therefore trusted over the mesh, and the mesh is used only for pictures and for downstream tools.

Two findings are locked into the test suite so they cannot quietly disappear. One test asserts that a design with no terminal propellant reproduces the validated two-phase motor exactly. Another asserts that an unpowered vertical dive cannot reach Mach 1.1, which is the infeasibility of section 1.2.

6. Fidelity and limitations

This is a Class-I conceptual tool. It is useful for choosing between configurations. It is not a substitute for computational fluid dynamics, for structural analysis or for six-degree-of-freedom flight mechanics.

The vehicle is invented. Its requirements correspond to no real programme. Section 1.1 shows that they were not even self-consistent when written.

6.1 The largest unquantified optimism

Nozzle flow is ideal. The model assumes frozen composition, constant specific-heat ratio and single-phase flow. It has no two-phase alumina loss, no divergence loss, no combustion inefficiency and no throat erosion. Real delivered specific impulse for this propellant class is typically 3 to 7 percent lower. **That penalty was not applied, because its magnitude could not be sourced.** Range and impact Mach are therefore optimistic by an amount that has not been quantified. This is the single largest known optimism in the result.

6.2 The fin fidelity mismatch

The fins are constant-thickness tapered plates in nTop. The aerodynamic model assumes a symmetric double wedge. The two do not agree. The plate has about twice the volume of a diamond of the same maximum thickness, so the fin mass is conservative by roughly 4 kg out of 43 kg of structure. The wave drag is computed for a section the geometry does not have.

An exact double wedge was attempted and abandoned for a specific reason. For a swept tapered panel the leading edge, the ridge and the trailing edge are three skew lines. The wedge faces are therefore hyperbolic paraboloids. The available loft operation interpolates signed-distance fields rather than boundaries, so it rounds the diamond corners: it returned 82 percent of the exact panel volume and 73 percent of its area. No shear or non-uniform-scale transform was available to build the correct surface.

6.3 Every value that is a guess

Every empirical constant in the code is registered with its source. A value that is a guess must say so in its source text, and a test asserts that. The table lists every registered source whose text admits a guess or a modelling choice.

Registered name	Why it is not a measured value
aero_nozzle_exit_fraction_GUESS	GUESS. Used only when `area_nozzle_exit` is None and `power_on=True`: the nozzle exit is assumed to occupy $A_e/A_{base} = 0.50$. This is a placeholder for the real value, which WP3's motor model must supply. It changes only the powered-base drag relief.
aero_protuberance_GUESS	GUESS. A single lumped allowance for antennas, umbilicals, fin-root fairings, lugs, joints and surface imperfections, taken as 5 percent of the sum of the other zero-lift components. There is no correlation behind this number. It appears in the breakdown as 'CD_protuberance_GUESS' so it can never be mistaken for a computed quantity.
bulkhead_allowance	GUESS: four ring bulkheads, each a full-diameter disc of 40 percent of wall thickness. Not from a source; a placeholder for joint, rail and lug hardware
case_minimum_gauge	Manufacturing minimum gauge, not a strength requirement. Hoop stress alone gives a sub-millimetre wall at this chamber pressure, which cannot be filament wound or machined. Floors of 1.8 mm (composite, roughly 3 to 4 wound plies) and 1.2 mm (metallic) are taken as engineering minimums. GUESS: no public minimum-gauge standard was found, but without the floor the model returns a motor mass fraction of 0.95, well above the 0.80 to 0.92 band for real tactical motors

Registered name	Why it is not a measured value
insulation_thickness	Sutton and Biblarz, Rocket Propulsion Elements, 9th ed., Chapter 14: EPDM chamber insulation 1 to 5 mm for short-burn tactical motors; 2.5 mm taken here as a mid-range modelling choice
prop.burn_rate_coefficient	MODELLING CHOICE inside a cited range: reference burn rate 10.0 mm/s at 1000 psia. Sutton Table 12-1 (HTPB/AP/Al) gives 0.25 to 3.0 in/s = 6.35 to 76.2 mm/s at 1000 psia, and the Ch.12 text on p.476 states most composite propellants burn at 7 to 20 mm/s. The coefficient a follows from $a = r_{ref} / p_{ref}^n$.
prop.case_safety_factor	MODELLING CHOICE: safety factor 1.5 applied to material yield strength in the thin-wall hoop-stress case sizing ($\sigma = p * r / t$), plus a 1.0 mm minimum practical gauge. 1.5 is the conventional aerospace pressure-vessel factor; it was not taken from a specific standard in this session.
prop.gamma	ASSUMPTION (not tabulated, treat as a guess inside a physically bounded band): ratio of specific heats of the combustion products taken as 1.20, the value normally used for aluminized AP/HTPB exhaust. Cross-checked: with $\gamma = 1.20$ the cited Isp (262.5 s) and flame temperature (3440 K) imply a mean exhaust molar mass of about 26 kg/kmol, inside the 25 to 30 kg/kmol band of Sutton Figure 12-3 for HTPB-based composite propellant at 68 atm. A +/- 0.02 change in γ moves the derived c^* by about 1.5 %.
prop.ideal_nozzle	APPROXIMATION: frozen-composition, constant- γ , single-phase isentropic nozzle flow. No two-phase Al_2O_3 loss, no divergence loss, no combustion efficiency and no throat erosion. Reported Isp is therefore an ideal value; real delivered Isp for a tactical motor of this class is typically 3 to 7 % lower (magnitude of that correction NOT sourced here, so it is not applied).
prop.igniter_mass	GUESS: igniter and safe-arm mass taken as 0.5 % of total propellant mass.
prop.insulation_thickness	GUESS: 3.0 mm of EPDM internal insulation on the case wall and domes. Real insulation is sized from local gas-side heat flux and exposure time, which is out of scope (SPEC.md section 8). Density from MATERIALS['insulation_epdm'].
prop.motor_mass_fraction	CROSS-CHECK CORRELATION, quoted band only, treat the endpoints as a guess: tactical solid rocket motors have propellant mass fractions $m_p / (m_p + m_{inert})$ of roughly 0.80 to 0.91. This band was not verified against a primary source in this session; it is used only to bound the physics-route inert mass, never to replace it silently.
prop.neutral_burning	APPROXIMATION: neutral (constant) burning is assumed within each phase. The burn-rate law is closed at the MEAN web position, which makes propellant mass, web thickness, mass flow and burn time mutually consistent ($w = V_p / A_{b_mean}$, $t_b = w / r$, $\dot{m} = \rho * A_{b_mean} * r$). No burnback simulation is performed, so the reported grain dimensions are volume- and area-equivalent, not a manufacturing drawing.
prop.nozzle_mass_model	GUESS-LEVEL geometric estimate: conical nozzle, 15 deg half angle, 8.0 mm wall, bulk graphite / carbon-carbon density 1800 kg/m ³ (handbook range 1.7 to 1.9 g/cm ³ , not verified against a datasheet in this session), multiplied by 1.5 to allow for the convergent section, throat insert and attach ring. This estimate is known to be optimistic; compare against SOURCES['prop.motor_mass_fraction'].
prop.ramp_times	MODELLING CHOICE, not physics: ignition rise 0.05 s (specified by the WP3 task statement), boost-to-sustain blend 0.10 s, tail-off 0.20 s. The ramps are mass-conserving by construction so total impulse is unchanged.

Registered name	Why it is not a measured value
prop.separation_criterion	Summerfield flow-separation criterion, separation when p_e / p_a is below about 0.4 (Summerfield, Foster and Swan, 1954; the 0.4 figure is the commonly quoted approximation and was NOT verified against the primary paper in this session). Used only to raise a warning, never to change a force.
prop.terminal_grain	MODELLING CHOICE: the terminal segment is an internal-burning tubular grain, the same geometry and the same mean-web closure as the boost segment, because an end burner at this bore cannot supply the burning area a multi-kilonewton terminal thrust needs at a credible chamber pressure. Its length counts towards the SPEC.md section 4 grain L/D limit of 1.0 to 8.0.
prop.terminal_pulse	MODELLING CHOICE with one guess: the terminal pulse is a third grain segment in the same case, ignited on command rather than on a timer, which is how real dual-pulse tactical motors work (an inter-pulse bulkhead separates the charges and a second igniter fires the aft one). Its default chamber pressure is $1.0 \times \text{DesignVector.p}_c$, that is the pressure the case is already sized for, so no new number is introduced. GUESS: the inter-pulse bulkhead, its insulation and the second igniter are NOT costed as separate inert mass; the terminal grain is charged only the same 0.5 % igniter fraction as the rest of the motor, so the reported inert mass is optimistic for a pulsed motor. Physical motivation: an unpowered terminal dive is terminal-velocity limited and cannot reach Mach 1.5 at sea level for any dive angle, so thrust in the endgame is the only way to meet SPEC R6.
prop.two_position_throat	MODELLING CHOICE: the motor is given a two-position throat (ejectable boost throat insert) so the boost and sustain phases can run at different throat areas with the same nozzle area ratio. Without it, a single throat sized for the boost thrust drives the sustain chamber pressure to about 0.13 MPa and Kn to about 22, which no composite propellant will sustain. Set two_position_throat=False to model one fixed throat instead.
radome_thickness	GUESS: 6 mm Pyroceram radome wall. No public thickness standard was found for a 0.35 m class radome; scaled from the requirement that wall thickness be an integer number of half-wavelengths at Ku band, which is not modelled here
traj.alpha_limit	MODELLING CHOICE: trim angle of attack is limited to 15 deg. Trim demands beyond that are recorded as an authority shortfall rather than flown, so the sizing loop can see them. 15 deg is a conventional tactical-rocket limit; it was not taken from a specific source in this session, so treat it as a guess.
traj.dive_entry	MODELLING CHOICE: SPEC.md section 2 does not define the coast-to-dive transition. Rule 'max_range' (default): hold altitude until level trim can no longer be held inside the alpha limit, or until Mach falls to 1.20, then dive at Requirements.gamma_terminal. Rule 'range': dive as soon as the current range plus the geometric dive range reaches Requirements.range_min. Rule 'terminal_boost': dive the moment the sustain phase ends, with no level coast at all, which keeps the most kinetic and potential energy for the endgame.
traj.gravity	Constant gravity $g = 9.80665 \text{ m/s}^2$ (standard gravity, CGPM 1901). MODELLING CHOICE: the inverse-square variation is neglected, which is under 0.4 % over the 0 to 12 km band of this mission and keeps the analytic validation cases in tests/test_trajectory.py exact.

Registered name	Why it is not a measured value
traj.guidance_gains	MODELLING CHOICE, no source: the altitude-hold and flight-path-angle autopilot gains (0.15 1/s outer, 2.0 1/s inner), the 200 m/s commanded climb rate cap, the 45 deg commanded climb angle cap and the 20 g commanded load factor cap are arbitrary. Guidance law design is an explicit non-goal (SPEC.md section 8); these gains only have to fly the SPEC.md section 2 profile smoothly. They do not appear in any reported performance number other than through the flown trajectory. The load factor cap matters for one thing only: it keeps the recorded CN_required history finite at the dive-entry command step, so the sizing loop can read it.
traj.terminal_boost_ignition	MODELLING CHOICE: SPEC.md section 2 has no terminal-boost phase, so the ignition trigger is invented here and exposed rather than hidden. Rule 'carry_to_impact' (default) lights the pulse when the altitude falls to $\text{terminal_ignition_margin} * V * \sin(\gamma_{\text{terminal}}) * t_{\text{terminal}}$, that is when the predicted remaining descent time equals the terminal burn time, so the pulse burns out at impact instead of above the ground. Rule 'dive_entry' lights it at dive entry. Rule 'never' leaves it unlit, which is what the two-phase regression case uses. The margin defaults to 1.0 and is a free parameter for the sizing loop: the prediction uses the CURRENT speed and the pulse accelerates the rocket, so a margin near 1.0 lights slightly late and the pulse is still burning at impact. That is the safe side; a margin above about 1.3 wastes propellant above the ground.

Table 16. All 25 registered values that are a guess, a modelling choice or an approximation. Source: runs/SV-1/provenance.json.

6.4 Other limitations

- The transonic drag rise is an interpolated blend between the subsonic value and the supersonic value at Mach 1.2. It is not physics. It is excluded from validation, and a test asserts that it underpredicts, so it can never be mistaken for a drag-rise model.
- The trajectory is a three-degree-of-freedom point mass on a flat earth with constant gravity. There is no roll, no sideslip and no autopilot.
- Guidance gains, the angle-of-attack limit and the dive-entry rule are arbitrary. Guidance design is out of scope. The load cap exists only to keep the recorded commanded normal force finite at the dive-entry step.
- Structural sizing is wall thickness times density, plus a hoop-stress check on the motor case with a minimum-gauge floor. There is no buckling, no flutter and no thermal analysis.
- The inter-pulse bulkhead, its insulation and the second igniter are not costed for the terminal pulse, so the pulsed motor inert mass is optimistic.
- The grain geometry is neutral-burning within each phase and closed at the mean web. There is no burnback simulation, so the reported grain dimensions are equivalent values rather than a drawing.
- The block universe vendored for notebook authoring is slightly stale against the installed nTop builds. Three specific mismatches were found and worked around.

7. How to reproduce this

Every artefact in this report comes from the commands below. The analysis writes JSON and CSV. The figure scripts read those files. This document reads them too, so no number is typed in twice.

```
.venv/Scripts/python.exe -m pytest tests -q
.venv/Scripts/python.exe run_sv1.py --stage smoke
.venv/Scripts/python.exe run_sv1.py --stage size
.venv/Scripts/python.exe run_sv1.py --stage doe --doe-scale full
.venv/Scripts/python.exe -m rocketgen.report.build_report
```

Component	Version or value
Python	3.11.15
numpy	1.26.4
scipy	1.13.1
SUAVE	2.5.2, vendored from github.com/suavecode/SUAVE
nTop Automate	5.53.2 installed, 5.54.0 development build
Drag calibration factor	1.171
Latin hypercube seed	20260817

Table 17. Environment. numpy and scipy are pinned below their current major versions because SUAVE 2.5.2 does not run on either.

One warning about the toolchain is worth recording. The nTop command-line tool returns exit code 72 when a block fails, not on success. Published guidance states the opposite. A notebook given a negative radius returned 72 and wrote no output at all. Success must therefore be judged on the expected files existing and being non-empty, never on the return code alone.